2 Functions and graphs

Week 2 · Worksheet

Name _____ Date _____

Exercises

Show your work. Each exercise has a hint at the end: try without it first.

1. Evaluate a formula Warm-up · Lesson 1

For $g(x) = 3x^2 - x + 2$, find $g(0)$, $g(-1)$, $g(2)$, and $g(t - 1)$.

Hint: Write $(t - 1)$ everywhere you see x, then expand $(t - 1)^2$.

2. Function or not? Warm-up · Lesson 1

Which rules are functions? (a) each phone → its serial number; (b) each city → the bus lines that stop there; (c) each day of 2026 → that day's highest temperature in your town; (d) each temperature → the day it was recorded.

Hint: Ask: can one input have two outputs?

3. A taxi ride Practice · Lesson 1

A taxi charges 4 to start plus 1.5 per km. Write the cost $C(d)$ for d km, find $C(12)$, and find the distance that costs 31.

Hint: $C(d) = 4 + 1.5d$. For the last part, solve $C(d) = 31$.

4. Is y a function of x? Practice · Lesson 1

Decide whether each equation defines y as a function of x: (a) $y^2 = x + 9$; (b) $y^3 = x$; (c) $2x - 5y = 10$.

Hint: Try to solve for y. Does a $\pm$ appear?

5. Four quick domains Warm-up · Lesson 2

Find the domain: (a) $f(x) = 5x^3 - x + 2$; (b) $g(x) = 3/(x^2 - 25)$; (c) $h(x) = \sqrt{(2x + 8)}$; (d) $p(x) = \log(7 - x)$.

Hint: Which of the three restrictions appears in each one?

6. A rational function Practice · Lesson 2

Find the domain of $f(t) = (t^2 + 1)/(t^2 + 2t - 15)$.

Hint: Factor the denominator.

7. A root of a quadratic Practice · Lesson 2

Find the domain and range of $r(x) = \sqrt{(x^2 - x - 6)}$.

Hint: Solve $(x - 3)(x + 2) \geq 0$ with a sign table.

8. A poster with a fixed frame Challenge · Lesson 2

A rectangular poster has a perimeter of 40 cm. With width w, its area is $A(w) = w(20 - w)$. Give the domain that makes sense and the range.

Hint: Both sides, w and 20 - w, must be positive. The vertex gives the largest area.

9. Even, odd, or neither Warm-up · Lesson 3

Classify: (a) $f(x) = 2x^6 - x^2$; (b) $g(x) = x^5 - 4x$; (c) $h(x) = x^3 - x + 2$; (d) $k(x) = |x| + x^2$.

Hint: Replace x by -x. Even powers and $|x|$ do not change; odd powers change sign.

10. Up, then down Practice · Lesson 3

Where is $f(x) = -x^2 + 4x + 1$ increasing and where decreasing? What is its maximum?

Hint: The vertex is at $x = -b/2a$.

11. A line that falls Practice · Lesson 3

For which values of m is $g(x) = (5 - 2m)x - 7$ decreasing?

Hint: The slope must be negative.

12. A day of phone battery Challenge · Lesson 3

A phone shows 100% at 8:00 and drains steadily to 40% at 14:00. A loose cable keeps it at 40% until 15:00, and then it charges to 90% at 16:00. Give the intervals where the battery level $B(t)$ increases, decreases, and is constant, and its range on $[8, 16]$.

Hint: Sketch it: three straight pieces.

13. A line through two points Warm-up · Lesson 4

Find the linear function with $f(-3) = 4$ and $f(3) = -2$.

Hint: $m = (-2 - 4)/(3 - (-3))$.

14. A gym membership Warm-up · Lesson 4

A gym charges a 25 sign-up fee plus 18 per month. Write the total cost $C(m)$ after m months. After how many months have you paid 241?

Hint: The fee is the intercept and the monthly price is the slope.

15. Asymptote or hole? Practice · Lesson 4

For $f(x) = (x + 4)/(x^2 - 16)$, give the domain, any holes, and the asymptotes.

Hint: $x^2 - 16 = (x - 4)(x + 4)$. Which factor cancels?

16. Graph a root function Practice · Lesson 4

Graph $g(x) = 2 - \sqrt{(x + 4)}$: domain, starting point, a table of values, and range.

Hint: Choose x values that make $x + 4$ a perfect square: -4, 0, 5, 12.

17. Trap a zero Challenge · Lesson 4

Show that $x^3 + x - 5 = 0$ has a solution between 1 and 2, then narrow it to an interval of length 1/2.

Hint: Evaluate at 1, 2, and 1.5 and look at the signs.

18. Pick the right piece Warm-up · Lesson 5

$g(x) = -x$ if $x \le 0$, x^2 if $0 < x < 3$, and 9 if $x \ge 3$. Find $g(-4)$, $g(0)$, $g(2)$, $g(3)$, and $g(10)$.

Hint: 0 belongs to the first piece and 3 to the last.

19. An electricity bill Practice · Lesson 5

Electricity costs 0.10 per kWh for the first 100 kWh and 0.15 for each kWh above 100. Write the bill $B(k)$ as a piecewise function and find $B(80)$ and $B(160)$.

Hint: Above 100 kWh you pay 10 for the first 100 plus 0.15 for the rest.

20. A shifted V Practice · Lesson 5

Write $|3x + 6|$ without bars, then find the vertex, zeros, and range of $y = |3x + 6| - 3$.

Hint: The inside is zero at $x = -2$.

21. Range of a three-piece function Challenge · Lesson 5

$h(x) = 3$ if $x \le -2$, $x^2 - 1$ if $-2 < x < 2$, and $-x + 5$ if $x \ge 2$. Find the domain and range, and say where the graph jumps.

Hint: Find the outputs of each piece on its own interval, then join them.

22. Describe the move Warm-up · Lesson 6

Describe each graph starting from the basic one: (a) $y = x^2 - 5$; (b) $y = (x + 4)^2$; (c) $y = |x - 1| + 2$; (d) $y = -\sqrt{x}$; (e) $y = \sqrt{(-x)}$.

Hint: Inside the function: horizontal, opposite sign. Outside: vertical, same sign.

23. Build the formula Practice · Lesson 6

Start with $y = |x|$. Move it 2 left, stretch it vertically by 3, and move it 6 down. Write the formula and find its zeros.

Hint: Left 2 means $x + 2$ inside.

24. Combine two functions Practice · Lesson 6

For $f(x) = x^2 - 4$ and $g(x) = x + 2$, find $f + g$, $f \cdot g$, and f/g with its domain.

Hint: $x^2 - 4 = (x - 2)(x + 2)$.

25. A bakery breaks even Challenge · Lesson 6

A bakery sells cookies at 4 each, so $R(q) = 4q$, and its costs are $C(q) = 150 + 1.5q$ per day. Find the profit function and the break-even quantity.

Hint: $P = R - C$, and break-even means $P = 0$.

26. Both orders Warm-up · Lesson 7

For $f(x) = x^2 - 3$ and $g(x) = 4x + 1$, find $f(g(x))$, $g(f(x))$, and $f(g(-1))$.

Hint: In $f(g(x))$, replace x in f by the whole $4x + 1$.

27. Domain of a composition Practice · Lesson 7

For $f(x) = \sqrt{(x - 2)}$ and $g(x) = x^2 + 1$, find $f(g(x))$ and $g(f(x))$ with their domains.

Hint: $g(f(x))$ simplifies, but x must still be allowed into f first.

28. Undo a line Warm-up · Lesson 7

Find the inverse of $f(x) = 7 - 2x$ and check it.

Hint: $y = 7 - 2x$; solve for x.

29. A spreading ripple Practice · Lesson 7

A stone falls into a pond and the ripple's radius grows as $r(t) = 30t$ cm after t seconds. The circle's area is $A(r) = \pi r^2$. Write the area as a function of time and evaluate it at $t = 2$.

Hint: You want $A(r(t))$.

30. Inverse after a restriction Challenge · Lesson 7

Find the inverse of $f(x) = (x - 4)^2 + 3$ for $x \geq 4$, with its domain and range.

Hint: Since $x \geq 4$, $x - 4$ is the positive square root.

31. Convert the angles Warm-up · Lesson 8

Convert $150°$ and $225°$ to radians, and $3\pi/2$ and 2 rad to degrees.

Hint: Degrees → radians: × $\pi/180$. Radians → degrees: × $180/\pi$.

32. Read the wave Warm-up · Lesson 8

Give the amplitude, period, and range of $y = 5\sin(4x) + 2$.

Hint: $A = 5$, $B = 4$, $D = 2$.

33. Evaluate with exact values Practice · Lesson 8

For $g(x) = \cos^2 x + 3\sin x$, find $g(\pi/2)$ and $g(\pi/3)$ exactly.

Hint: $\cos(\pi/3) = 1/2$ and $\sin(\pi/3) = \sqrt{3}/2$.

34. Where two waves meet Practice · Lesson 8

How many solutions does sin x = cos x have in $[0, 4\pi]$? List them.

Hint: Divide by cos x: tan x = 1, which repeats every π.

35. Hours of daylight Challenge · Lesson 8

In one city, the daylight on day t of the year is about $D(t) = 12 + 2.5 \sin(2\pi(t - 80)/365)$ hours. Find the longest and shortest day lengths, the period, and the day with the most light.

Hint: The sine reaches 1 when its angle is π/2.

36. A substitution identity Challenge · Lesson 8

If $x = 3 \cos t$ with t in the first quadrant, show that $\sqrt{(9 - x^2)} = 3 \sin t$.

Hint: Use $1 - \cos^2 t = \sin^2 t$, and decide the sign of sin t.

37. Logs in your head Warm-up · Lesson 9

Evaluate: $\log_5 125$, $\log_2(1/8)$, $\ln(e^4)$, log 0.01, and 10^(log 7).

Hint: Ask which power of the base gives the number.

38. Same base Practice · Lesson 9

Solve $4^x = 8^{(x - 1)}$.

Hint: Write both sides as powers of 2.

39. Bacteria in a dish Practice · Lesson 9

A culture starts with 500 bacteria and triples every 2 hours. Write N(t), find N(6), and find when it reaches 40,500.

Hint: N(t) = 500·3^(t/2). For the last part, divide by 500 first.

40. Expand a logarithm Practice · Lesson 9

Expand $\ln(x^3\sqrt{y} / e^2)$ using the laws of logarithms.

Hint: √y = y^(1/2), and ln e² = 2.

41. When does it double? Challenge · Lesson 9

2,000 is invested at 3.5% compounded continuously: A(t) = 2000e^(0.035t). How long until it doubles?

Hint: Solve e^(0.035t) = 2 with ln.

42. A log equation with a trap Challenge · Lesson 9

Solve $\log_2 x + \log_2(x - 2) = 3$.

Hint: Combine into one log, rewrite as a power, and check both candidates.

Quick check

Circle one answer.

43. Which rule is NOT a function?

A) Each person → their date of birth B) Each date → the people born that day

C) Each product → its price today

44. What is the range of $f(x) = 5 - x^2$?

A) $(-\infty, 5]$ B) $[5, \infty)$ C) $\mathbb{R}$

45. A café's daily profit is $P(t) = -2t^2 + 24t - 40$ (t = hours after opening). On which interval is profit rising?

A) [0, 6] B) [6, 12] C) [0, 32]

46. A student writes: "$(x^2 - 9)/(x - 3) = x + 3$, so its graph is the whole line $y = x + 3$." What is wrong?

A) The point (3, 6) is missing: $x = 3$ is not in the domain B) It simplifies to $x - 3$, not $x + 3$

C) There is a vertical asymptote at $x = 3$

47. A parking garage charges 4 for the first hour or part of it, then 2 for each extra started hour. What do you pay for 3 h 10 min?

A) 10 B) 8 C) 10.33

48. Where is the vertex of $y = -(x + 2)^2 - 3$?

A) (-2, -3), opening down B) (2, -3), opening down C) (-2, 3), opening up

49. Solve $3e^{(2x)} = 12$.

A) $x = \ln(4)/2 \approx 0.69$ B) $x = \ln(12)/6$ C) $x = 2$

50. A student simplifies $\log(x + 10)$ as $\log x + 1$. Where is the error?

A) Logs split products, not sums: $\log(10x) = \log x + 1$, but $\log(x + 10)$ does not split B) log 10 is 10, not 1

C) There is no error

Answer key

1. $g(0) = 2$, $g(-1) = 3 + 1 + 2 = 6$, $g(2) = 12 - 2 + 2 = 12$, $g(t - 1) = 3(t^2 - 2t + 1) - (t - 1) + 2 = 3t^2 - 7t + 6$. Check: $t = 0$ gives $6 = g(-1)$.
2. (a) and (c) are functions. (b) is not: a city has many bus lines. (d) is not: the same temperature can happen on several days.
3. $C(12) = 4 + 18 = 22$. $4 + 1.5d = 31 \rightarrow 1.5d = 27 \rightarrow d = 18$ km.
4. (a) No: $x = 0$ gives $y = 3$ and $y = -3$. (b) Yes: $y = \sqrt[3]{x}$, one cube root for each x. (c) Yes: $y = (2x - 10)/5$, a line.
5. (a) $\mathbb{R}$; (b) $\mathbb{R} \setminus \{-5, 5\}$; (c) $2x + 8 \geq 0 \rightarrow [-4, \infty)$; (d) $7 - x > 0 \rightarrow (-\infty, 7)$.
6. $t^2 + 2t - 15 = (t + 5)(t - 3)$, so the domain is $\mathbb{R} \setminus \{-5, 3\}$.
7. Domain $(-\infty, -2] \cup [3, \infty)$. The radicand takes every value ≥ 0 there, so the range is $[0, \infty)$.
8. Domain $(0, 20)$. The vertex is at $w = 10$ with $A = 100$, and near the ends the area approaches 0: range $(0, 100]$.
9. (a) even; (b) odd; (c) neither ($h(-x) = -x^3 + x + 2$); (d) even.
10. Vertex $x = 2$, $f(2) = 5$. Increasing on $(-\infty, 2]$, decreasing on $[2, \infty)$, maximum 5.
11. $5 - 2m < 0 \rightarrow m > 5/2$.
12. Decreasing on $[8, 14]$, constant on $[14, 15]$, increasing on $[15, 16]$. Lowest level 40%, highest 100%: range $[40, 100]$.
13. $m = -6/6 = -1$. $4 = -1 \cdot (-3) + b$ gives $b = 1$, so $f(x) = -x + 1$.
14. $C(m) = 18m + 25$. $18m + 25 = 241 \rightarrow 18m = 216 \rightarrow m = 12$ months.
15. Domain $\mathbb{R} \setminus \{-4, 4\}$. For $x \neq -4$, $f(x) = 1/(x - 4)$: a hole at $(-4, -1/8)$, a vertical asymptote $x = 4$, and a horizontal asymptote $y = 0$.
16. Domain $[-4, \infty)$, start $(-4, 2)$. Table: $(-4, 2)$, $(0, 0)$, $(5, -1)$, $(12, -2)$. It always decreases; range $(-\infty, 2]$.
17. $f(1) = -3 < 0$ and $f(2) = 5 > 0$, so a zero lies in $(1, 2)$. $f(1.5) = 3.375 + 1.5 - 5 = -0.125 < 0$, so it lies in $(1.5, 2)$.
18. $g(-4) = 4$, $g(0) = 0$, $g(2) = 4$, $g(3) = 9$, $g(10) = 9$.
19. $B(k) = 0.10k$ if $0 \leq k \leq 100$, and $10 + 0.15(k - 100)$ if $k > 100$. $B(80) = 8$, $B(160) = 10 + 9 = 19$.
20. $|3x + 6| = 3x + 6$ if $x \geq -2$ and $-3x - 6$ if $x < -2$. Vertex $(-2, -3)$. Zeros: $|3x + 6| = 3$ gives $x = -1$ or $x = -3$. Range $[-3, \infty)$.
21. Domain $\mathbb{R}$. Outputs: $\{3\}$, then $[-1, 3)$ for the middle, then $(-\infty, 3]$ for the last piece. Range $(-\infty, 3]$. No jumps: at $x = -2$ the middle piece approaches $3 = h(-2)$, and at $x = 2$ it approaches $3 = h(2)$.
22. (a) down 5; (b) left 4; (c) right 1, up 2; (d) flipped over the x-axis; (e) flipped over the y-axis (domain $x \leq 0$).
23. $y = 3|x + 2| - 6$. Zeros: $|x + 2| = 2$, so $x = 0$ or $x = -4$. Vertex $(-2, -6)$.
24. $f + g = x^2 + x - 2$; $f \cdot g = x^3 + 2x^2 - 4x - 8$; $f/g = x - 2$, but only for $x \neq -2$: domain $\mathbb{R} \setminus \{-2\}$.
25. $P(q) = 4q - (150 + 1.5q) = 2.5q - 150$. $P = 0$ at $q = 60$ cookies; above that the bakery makes money.
26. $f(g(x)) = (4x + 1)^2 - 3 = 16x^2 + 8x - 2$; $g(f(x)) = 4(x^2 - 3) + 1 = 4x^2 - 11$; $f(g(-1)) = f(-3) = 6$.
27. $f(g(x)) = \sqrt{(x^2 - 1)}$, domain $(-\infty, -1] \cup [1, \infty)$. $g(f(x)) = (x - 2) + 1 = x - 1$, but only for $x \geq 2$, because f must accept x first.
28. $x = (7 - y)/2$, so $f^{-1}(x) = (7 - x)/2$. Check: $f(f^{-1}(x)) = 7 - (7 - x) = x$.
29. $A(r(t)) = \pi(30t)^2 = 900\pi t^2$. At $t = 2$: $3600\pi \approx 11{,}310$ cm^2.
30. $y - 3 = (x - 4)^2 \rightarrow x = 4 + \sqrt{(y - 3)}$. $f^{-1}(x) = 4 + \sqrt{(x - 3)}$, domain $[3, \infty)$ (the range of f) and range $[4, \infty)$ (the domain of f).
31. $150° = 5\pi/6$, $225° = 5\pi/4$, $3\pi/2 = 270°$, 2 rad $\approx 114.6°$.
32. Amplitude 5, period $2\pi/4 = \pi/2$, range $[2 - 5, 2 + 5] = [-3, 7]$.

33. $g(\pi/2) = 0 + 3 \cdot 1 = 3$. $g(\pi/3) = 1/4 + 3\sqrt{3}/2 \approx 2.85$.

34. $x = \pi/4 + n\pi$: $\pi/4$, $5\pi/4$, $9\pi/4$, $13\pi/4$. Four solutions ($17\pi/4$ is beyond 4π).

35. Longest 14.5 h, shortest 9.5 h, period 365 days. $2\pi(t - 80)/365 = \pi/2$ gives $t - 80 = 91.25$, so $t \approx 171$, around June 20.

36. $\sqrt{(9 - 9\cos^2 t)} = \sqrt{(9 \sin^2 t)} = 3|\sin t| = 3 \sin t$, because $\sin t > 0$ in the first quadrant.

37. 3, -3, 4, -2, and 7.

38. $2^{(2x)} = 2^{(3x - 3)}$, so $2x = 3x - 3$ and $x = 3$. Check: $4^3 = 64 = 8^2$.

39. $N(6) = 500 \cdot 3^3 = 13{,}500$. $3^{(t/2)} = 81 = 3^4$, so $t/2 = 4$ and $t = 8$ hours.

40. $3 \ln x + (1/2) \ln y - 2$.

41. $0.035t = \ln 2$, so $t = \ln 2 / 0.035 \approx 19.8$ years.

42. $\log_2(x(x - 2)) = 3 \rightarrow x^2 - 2x = 8 \rightarrow (x - 4)(x + 2) = 0$. $x = -2$ makes $\log_2 x$ undefined, so only $x = 4$.

43. **B** — Many people are born on the same day, so one date would have many outputs.

44. **A** — $x^2 \geq 0$, so $5 - x^2 \leq 5$, and every value below 5 is reached.

45. **A** — The parabola opens down with vertex $t = -24/(2 \cdot (-2)) = 6$, so it rises until $t = 6$ ($P(6) = 32$) and falls after.

46. **A** — The factor $x - 3$ cancels, which leaves a hole at $(3, 6)$, not an asymptote.

47. **A** — 3 h 10 min starts 4 hours: $4 + 3 \cdot 2 = 10$. A step function never gives fractions of a step.

48. **A** — $x + 2$ inside moves left 2; -3 outside moves down 3; the minus in front flips it.

49. **A** — Divide by 3 first: $e^{(2x)} = 4$, so $2x = \ln 4$ and $x = \ln(4)/2 = \ln 2 \approx 0.69$.

50. **A** — $\log(xy) = \log x + \log y$ is about products. Try $x = 10$: $\log 20 \approx 1.30$, but $\log 10 + 1 = 2$.